# Sandra Yang

aleyang@stanford.edu | sandrayang.com

## Education

**Stanford University** *B.S. Computer Science + Mathematics* Jun. 2028

- Coursework: Machine Learning, Deep Learning, Reinforcement Learning, Operating Systems, Algorithms, Probability, Statistical Inference, Stochastic Processes, Real Analysis, Combinatorics

## Experience

**Ramp** *Software Engineer Intern* Jun. 2026 – Sep. 2026

- Early engineer on Ramp Router, an LLM gateway that cuts inference costs ~40%: helped design and build the initial gateway, onboarding, and attribution, and wrote the first intelligent routing strategies (semantic difficulty-based model selection).
- Co-ran BugBasher, a four-week experiment giving an autonomous agent (Devin) a Ramp card, phone, and email to run a B2B business from a cold start; built the heartbeat automations, shared "brain" memory store, and voice pipeline (Twilio, ElevenLabs, Stripe) behind the first publicly documented AI-agent cold-start sale.

**Stanford AI Lab (SAIL)** *Undergraduate Researcher, advised by Tengyu Ma* 2026 – Present

- FuzzyDecisionBench (*Evaluating Open-Ended Decision-Making With LLM-as-Environment*, w/ Caroline Choi): a benchmark of LLM-simulated environments for open-ended sequential decision-making; 3 environments, 11 frontier models evaluated. Accepted to NeurIPS 2026 Verify-Agents Workshop; under review at ICLR 2027.
- Built the NFL head-coach environment (prompt + dynamics engine), the multi-seed probe orchestrator, the human-evaluation UI, and the project website.

**Andreessen Horowitz (a16z)** *New Media* Mar. 2026 – Jun. 2026

- Built application infrastructure for a16z fellowship programs end to end (intake, review, and tooling).
- Hosted *Monitoring the Situation*, a daily tech show on X.

**Khosla Ventures** *Technical Intern to Vinod Khosla* Jan. 2026 – Mar. 2026

- Built internal platform tooling and AI/ML pipelines for portfolio evaluation; worked directly with Vinod Khosla on infrastructure and investment. TypeScript, Drizzle ORM, PostgreSQL, AWS.

**Human Behavior (YC X25)** *Founding Engineer* Jun. 2025 – Sep. 2025

- First hire; took the core product and platform from 0 to 1 across frontend, backend, AI/ML, design, and product. TypeScript, Postgres, LangChain, React Native, Node.js.

**Jane Street** *First-Year Trading & Technology Program* Mar. 2025

**Harvard Visual Attention Lab** *Research Intern, Research Science Institute (RSI)* Jun. 2023 – Aug. 2023

- Algorithms extracting predictive signal from unskilled crowd data for breast cancer detection; adopted by a medical data-labelling company (+21% prediction accuracy).

**University of Virginia Biocomplexity Institute** *Algorithms Developer Intern* Aug. 2021 – Jul. 2023

- Hybrid parallel merge-sort algorithms (Python, OpenMPI, multiprocessing) with 34x speedups; co-authored preprint *Approaches to the Parallelization of Merge Sort in Python.*

## Projects

**LeMMA: Test-Time Meta-Learning for Theorem-Proving LLMs** | *PyTorch, vLLM, Lean 4*

- Bilevel-optimization RL algorithm (GRPO) that trains an LLM to generate and weight problem-specific synthetic lemmas at inference time, then adapts on them before proving. Beat DeepSeek-Prover-V2 by 43% on miniF2F.
- LoRA + mixed-mode bilevel differentiation for memory-efficient meta-gradients on GPU.

**Stanford Programming Contest (ProCo)** | *Co-Director, Problem-Writing Lead* | *Python, C++, Polygon*

- Wrote 15+ algorithmic problems (DP, graphs, combinatorics, math) for 300+ Bay Area students; built generators and validators and deployed via Polygon. Problem sets: 2025, 2026.

## Awards & Skills

**Awards**: USACO Platinum; 4x AIME Qualifier; 2x CyberPatriot National Finalist & Boeing Champion; NCSF National Cyber Scholar; Research Science Institute (RSI) 2023; PROMYS 2022; Canada/USA Mathcamp 2021

**Languages**: Python, C++, C, TypeScript, Java, JavaScript, HTML/CSS, Bash, SQL, Lean 4

**Data / ML**: PyTorch, vLLM, scikit-learn, LangGraph, NumPy, pandas, SciPy, OpenCV, Matplotlib

**Frameworks / Infra**: React, Next.js, Node.js, PostgreSQL, Supabase, Prisma, Drizzle, AWS, Vercel, CI/CD, Distributed Systems, Parallel Programming, OpenMPI